

M7: Homework

● Graded

Student

Jonathan C Elder

Total Points

50 / 50 pts

Question 1

7.1.1

5 / 5 pts

✓ - 0 pts Correct

Question 2

7.1.4

5 / 5 pts

✓ - 0 pts Correct

Question 3

7.2.2

5 / 5 pts

✓ - 0 pts Correct

Question 4

7.4.1

5 / 5 pts

✓ - 0 pts Correct

Question 5

7.4.2

5 / 5 pts

✓ - 0 pts Correct

Question 6

7.5.1

5 / 5 pts

✓ - 0 pts Correct

Question 7

7.6.1

5 / 5 pts

✓ - 0 pts Correct

Question 8

7.7.2

5 / 5 pts

✓ - 0 pts Correct

Question 9

7.8.1

5 / 5 pts

✓ - 0 pts Correct

Question 10

7.9.1

5 / 5 pts

✓ - 0 pts Correct

Question assigned to the following page: [1](#)

7.1 HW

7.1.1 a) total degree of $G = d(1) + d(2) + d(3) + d(4) + d(5) + d(6)$
 $= 2 + 3 + 2 + 1 + 5 + 1$

degree of $G = 14$

b) Neighbors of the vertex 5 = $\{1, 2, 3, 6, 4\}$

c) Degree of Vertex 6 = 1

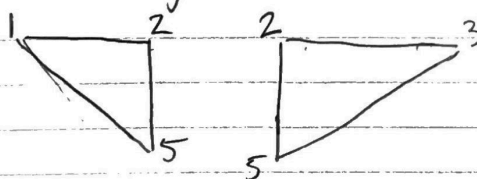
d) Vertices that are adjacent to vertex 3: $\{2, 5\}$

e) G is not a regular graph, because the degree of each vertex is not the same.
E.g. $d(1) = 2, \neq d(5) = 5$

f) K_3 is a subgraph of G in

2 ways.

The vertices are $\{1, 2, 3, 5\}$



g) K_4 is not a subgraph of G , because there is no way to complete a cycle w/ 4 edges.

Question assigned to the following page: [2](#)

7.1 HW Cont.

7.1.4 a) Number of edges = 3×4
 $= 12$ And $K_{3,4}$ is not a regular graph.

b) Number of edges $K_5 = \frac{5 \times (4/2)}{2}$
 $= 10$ And K_5 is a 4-regular graph.

c) What is the largest n such that $K_n = C_n$?

$$K_3 = C_3 \text{ So, } n=3$$

d) For what value of n is $Q_2 = C_n$?

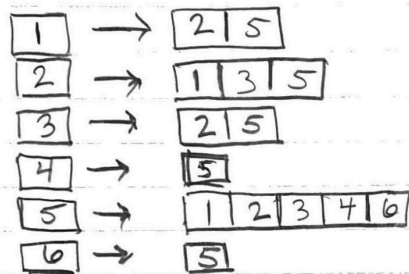
$$Q_2 = C_4 \text{ So, } n=4$$

e) Yes it would be a regular graph.
And each vertex would have " n " degree.
 $d(Q_n) = n$

Question assigned to the following page: [3](#)

7.2 HW

7.2.2 a) Adjacency List Representation of G



b) Matrix Representation of G

$$G = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{matrix}$$

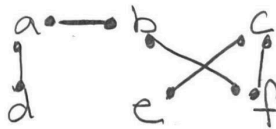
Questions assigned to the following page: [4](#) and [5](#)

7.4 HW

7.4.1 a) Four components: $\{a, b, c\}$, $\{d, e\}$, $\{g\}$, $\{f, h, i, j\}$

b) Since there are no edges and each vertex is connected to itself. $\{a\}$, $\{b\}$, $\{c\}$, $\{d\}$, $\{e\}$

c) This is one component as they are all connected.



7.4.2 a) The vertex connectivity is 1, because if vertex b is removed the graph becomes disconnected. The edge connectivity is 2, because the minimum degree of any vertex is 2.

b) The vertex connectivity is 1, because if vertex g is removed the graph becomes disconnected. The edge connectivity is 1, because the minimum degree of any vertex is 1.

c) The vertex connectivity is 3, because there must be 3 removed vertices before the graph becomes disconnected. The edge connectivity is also 3, because there must be 3 edges removed before the graph becomes disconnected.

Question assigned to the following page: [6](#)

7.5 HW

7.5.1 a) $a=4$ $e=2$
 $b=4$ $f=4$
 $c=4$
 $d=4$

euler circuit: $\langle a, b, a, d, c, f, b, e, d, f, a \rangle$

b) $a=4$ $d=4$
 $b=3$ $e=2$
 $c=2$ $f=3$

euler circuit: $\langle b, a, c, d, a, f, b, e, d, f \rangle$

c) $a=3$ $d=3$
 $b=3$ $e=1$
 $c=4$ $f=4$
 $g=2$

The euler circuit is not possible with this graph, as there are more than 2 vertices having an odd degree.

d) $a=4$ $d=4$
 $b=4$ $e=2$
 $c=4$ $f=3$
 $g=2$

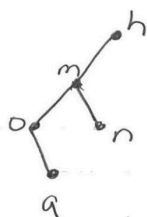
euler circuit: $\langle a, b, c, a, d, c, g, f, b, e, d, f, a \rangle$

Question assigned to the following page: [7](#)

7.6 HW

- 7.6.1
- a) Ancestors of vertex n : $\{m, h, L, p\}$
 - b) descendants of vertex "i" : $\{f, g, e, d, a, b, c\}$
 - c) Leaves : $\{j, q, e, k, g, a, b, c, n\}$
 - d) level of vertex d : 4
 - e) height of tree : 6
 - f) Level 4 vertices : $\{m, e, d\}$

g) Subtree of h :



h) Siblings of vertex "i" : $\{L, k\}$

Question assigned to the following page: [8](#)

7.7 HW

7.7.2 a) Day is encoded as "11100111". $a \rightarrow 0$

$e \rightarrow 10$

$c \rightarrow 1100$

b) Candy is encoded as
"110001101110111".

$n \rightarrow 1101$

$d \rightarrow 1110$

$y \rightarrow 1111$

c) Decode "1110101101": Den

d) Decode "11100110110010": Dance

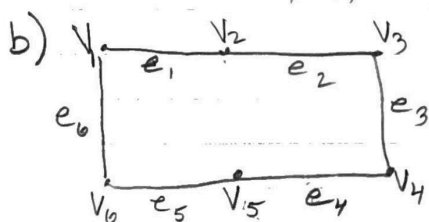
Question assigned to the following page: [9](#)

7.8 HW

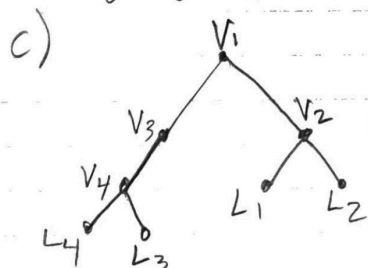
7.8.1 a) total degree = $\sum \deg(v)$

$$\begin{aligned} &= 2|E(G)| \\ &= 2 \cdot 6 \\ &= 12 \\ &\neq 14 \end{aligned}$$

So, there is no tree that has 7 vertices and a total degree of 14

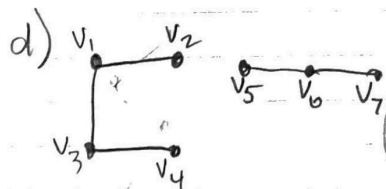


Yes, there exist a graph with 6 vertices and 6 edges.



4 internal vertices $\{v_1, v_2, v_3, v_4\}$
4 leaves $\{L_1, L_2, L_3, L_4\}$

Yes there exist a graph.



Yes there exist a graph with 7 vertices, 5 edges, and is acyclic.

e) $|E(G)| = n - 1$ $\deg(v_i) = 2$

$$\sum_{i=1}^n \deg(v_i) = 2|E|$$

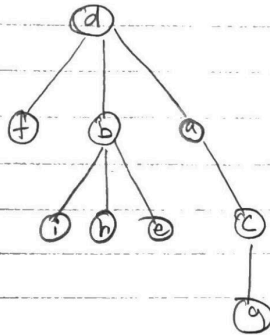
$$2n = 2(n-1)$$

← this is not true so there is no tree with degree of all vertices is 2.

Question assigned to the following page: [10](#)

7.9 HW

7.9.1



a) Post-order traversal of given graph : f, i, h, e, b, g, c, a, d

b) Pre-order traversal of given graph:

Pre-order: d, f, b, i, h, e, a, c, g